ASIAN INSTITUTE OF TECHNOLOGY

Faculty of Technology

Proposal Report on

Recommending Career Path for Masters' Graduates:
A Business Intelligence Analysis of Master's degree Pathways

Business Intelligence and Analytics (BIA)

SEMESTER: JANUARY 2026

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February 2026

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Introduction

In today's rapidly changing labor market, choosing a Master's degree has become a strategic decision rather than only an academic one. Automation, AI adoption, and shifting industry demands are reshaping job roles and creating uncertainty about which postgraduate specialisations will lead to stable employment Although Master's degrees are often viewed as improving employability, not all pathways offer the same long-term resilience.

At the same time, most career guidance available to students relies on anecdotal advice, historical assumptions, or short-term salary signals, rather than data driven analysis of employment scale, wage stability and occupational risk. This makes it difficult for graduates to confidently answer a key question: which career paths linked to a master's degree are most likely to remain stable soon?

This project addresses that gap by applying Business Intelligence and Analytics to integrate master's degree data, employment data, and occupation level risk indicators into an interactive dashboard. The project will also develop a **Graduate Career Stability Index** (GCSI) to compare career paths based on income strength (median annual salary), wage stability (salary dispersion, P90-P10 spread) and automation vulnerability. The final output will function as a decision support tool to help graduates choose more stable career directions.

Background and Problem Statement

Choosing a Master's degree has become a high-stakes decision for graduates due to rapid technological change, automation, and evolving labor market demands. Students frequently lack data-driven tools that can help them understand which career paths are likely to remain viable and stable. This project addresses this problem by developing a Business Intelligence (BI) system that evaluates the future career stability of common Master's degree pathways available. The focus is not on tuition cost or salary maximization, but on employment scale and wage stability, and long-term demand for the job.

Project Aim

This project aims to design and implement a Business Intelligence dashboard that forecasts the career stability of Master's graduates and recommends the most data driven stable career paths, in a very user-friendly dashboard. It might help the graduates or current Masters degree students decide what they want to become once they graduate and maybe will have flexible time to upgrade their skill based on the required skills for their job standing.

Project Objectives

- Identify common Master's degree categories offered worldwide.
- Map each Master's degree to realistic and data-driven career paths.
- Analyze employment scale and wage stability on each occupation.
- Estimate future career stability using a composite index.
- Develop an interactive BI dashboard to support graduate career decision-making

Scope of the Project

This project focuses on analyzing the relationship between commonly pursued Master's degree categories and their associated career paths, with the objective of assessing career stability. The scope of the analysis is global, using internationally available datasets to ensure consistency across education, labor market, and occupational risk data.

The project includes the identification and categorization of Master's degree programs using a worldwide dataset of postgraduate programs, which will be grouped into broad degree categories such as Computer Science, Business Administration, Mechanical Engineering, History and Nursing. The analysis is limited to widely recognized and commonly offered Master's degree pathways rather than niche or highly specialized programs.

Career paths associated with each degree category will be derived from employment and salary data accessible to graduates. Job market analysis will consider job titles, employment size, income strength. Salary maximization and individual employer-specific outcomes are outside the scope of this project.

To estimate long term career resilience, the project incorporates occupation level automation risk indicators as a proxy for vulnerability to technological disruption. These indicators will be used to support comparative analysis rather than precise prediction of future job displacement.

The final deliverable of the project is a single Business Intelligence dashboard that presents a master program related to career strength, ranked career stability scores and explanatory factors . The project does not aim to provide personalized career advice or definitive employment forecasts, but rather a data driven decision support framework to guide Master's graduates toward more stable career paths.

Data Sources

Dataset 1: Master's Degree Categories

Source: Masters portal Programs Dataset (Kaggle)
Link: https://www.kaggle.com/datasets/anasfullstack/mastersportal-programs/data?

This dataset provides a worldwide list of Master's degree programs, including program titles, universities, and countries. In this project, we'll use data available on the country "USA" since the occupation related data are only available for the USA and will be used to identify categories commonly offered master's degree pathways rather than individual institutions or niche programs.

The dataset will be cleaned and grouped into broad, recognizable degree categories such as Computer Science, Business Administration, Mechanical Engineering, History and Nursing. These degree categories form the starting point of the analysis, representing the educational pathways that graduates typically choose before entering the labor market.

The output of this dataset is a structured "Degree Categories" table, which serves as the foundation for mapping degrees to potential career paths in later stages of the project.

Dataset 2: Career Stability and Automation Risk Indicators

Source: Occupation, Salary, and Likelihood of Automation Dataset (Kaggle)
Link:
https://www.kaggle.com/datasets/andrewmvd/occupation-salary-and-likelihood-of-automation?

This dataset provides occupation-level information on employment, salary, and the estimated likelihood of automation for different job roles. In this project, it is used specifically as a proxy measure of long-term career vulnerability.

The dataset will be used to:

- Map identified degree program to corresponding occupation categories
- Assign an automation risk score to each categorized program.
- Convert automation risk into an "automation safety" indicator for comparative analysis

Because precise future automation outcomes cannot be predicted, this dataset is not used to forecast job loss directly. Instead, it supports relative comparison of career paths, helping to distinguish roles that are more resilient to automation from those that are more exposed.

This dataset contributes to the risk dimension of the Graduate Career Stability Index, complementing job demand and skill-based indicators.

How Two Datasets Work Together
Together, the three datasets enable an end-to-end analytical workflow:

- Dataset 1 defines what graduates study
- Dataset 2 estimates which career paths are more or less vulnerable in the long term

By integrating these datasets into a single Business Intelligence dashboard, the project produces a data-driven framework that evaluates and compares career stability across master's degree pathways.

Methodology

Step 1: Degree Classification

In the first step, Master's degree programs will be organized into broad degree categories. Individual program titles from the master's degree dataset will be cleaned to remove duplication and variations in naming. Programs will then be grouped based on their discipline and primary skill orientation, such as technical, analytical, managerial, or interdisciplinary.

For example, programs containing terms such as *Data Science*, *Artificial Intelligence*, *Analytics*, or *Machine Learning* will be grouped under a single "Data Science and AI" category. This approach ensures the analysis focuses on commonly pursued and recognizable Master's pathways, rather than individual universities or niche programs. The output of this step is a structured list of degree categories that serves as the foundation for the analysis.

Step 2: Career Path Mapping

In the second step, employment data will be analyzed to identify career paths associated with each master's degree category. Job titles from the occupation dataset will be standardized and job titles will be mapped into common career paths. For example, graduates from a Data Science and AI degree may be linked to career paths such as Data Scientist, Data Engineer, Machine Learning Engineer, BI Analyst, or Analytics Engineer. This ensures that career outcomes are derived from actual labor market data, rather than assumed or theoretical career trajectories.

The output of this step is a degree-to-career mapping that links each master's degree category to multiple realistic career paths.

Step 3: Graduate Career Stability Index (GCSI) Construction

In this step, a **Graduate Career Stability Index (GCSI)** will be constructed to summaries multiple stability factors into a single score. Each component is measured on a 0 to 100 scale, where higher values indicate greater stability.

GCSI formula:

This study proposes a Graduate Career Stability Index (GCSI) to compare the relative stability of career pathways associated with different Master's degree programs. Career stability is defined as the ability of graduates to access sufficient employment opportunities, maintain employability through transferable skills, receive relatively stable earnings, and remain resilient to automation risk.

The GCSI is constructed as a weighted composite index using four labor market dimensions:

- **Income Strength**, representing average annual salary of the occupation
- **Employment Scale**, capturing the overall size of employment in related occupations
- **Wage Stability**, measured through dispersion in occupational wages
- **Automation Safety**, indicating lower exposure to automation risk

All components are normalized to a 0–1 scale using min–max normalization to ensure comparability. Weights are assigned to reflect the relative importance of each dimension while avoiding redundancy between demand measures.

The index is calculated as:

$$GCSI = (0.35 \times \text{Income Strength}) + (0.25 \times \text{Employment Scale}) + (0.20 \times \text{Wage Stability}) + (0.20 \times \text{Automation Safety})$$

The resulting GCSI score ranges from 0 to 1, with higher values indicating comparatively more stable graduate career pathways. The index is intended for comparative analysis and provides a transparent framework rather than a precise prediction of individual outcomes.

Higher GCSI values indicate more stable career paths. Income Strength and Employment Scale are given higher weight because they reflect current labor market conditions and wage strength through the whole year, while wage stability and automation safety capture medium-term resilience. Automation risk is included as a supporting factor to reflect long-term vulnerability.

Step 4: Dashboard Development

In the final step, an interactive Business Intelligence dashboard will be developed using Power BI. The dashboard will allow users to select a master's degree category and view:

- Ranked career paths based on GCSI scores
- Master Degree Programs ranking
- Automation Probability varies widely across occupations
- Wages by major occupational group
- The key factors contributing to career stability

The dashboard is designed as a **decision-support tool**, translating complex labor market data into clear and actionable insights to help graduates choose more stable career directions.

Expected Outcomes

This project will deliver a structured, data driven approach for assessing the career stability of master's graduates by linking master's degree categories to labor market evidence. It will produce a Graduate Career Stability Index (GCSI) that combines employment scale, stable earnings, and remaining resilient to automation risk into a single comparable score for each career path. The project will also deliver an interactive Power BI dashboard that ranks career paths by stability, and explains the key drivers behind each score. Overall, the outputs will provide practical insights to help graduates choose more stable career directions and understand which skills to focus on for long term employability.

Tools and Technologies

Data Preparation:
Python will be used for initial data cleaning, filtering, and restructuring. Since Python can handle data cleaning, preparation, formatting, preprocessing, and feature engineering, EDA and preprocessing tasks will be performed in a Jupyter Notebook.

Visualization and Business Intelligence:
Power BI will be used to integrate all datasets and present the results through an interactive dashboard. Power BI enables dynamic filtering by degree category and career path, calculation of the Graduate Career Stability Index (GCSI), and clear visual communication of career stability scores and recommendations. The tool is selected for its suitability in building decisions to support dashboards and its ability to combine analytical measures with intuitive visualizations.

Conclusion

This project is significant because it applies Business Intelligence and analytics to a practical and high-impact problem: how postgraduate students can choose career paths that are more likely to remain stable. Rather than relying on informal advice or short-term trends, the project uses data to link master's degree pathways to real labor market demand, supporting more informed decision-making.

Academically, the project aligns strongly with Master's-level BIA learning outcomes by integrating multiple types of analytics. It includes descriptive analytics to summarize degree categories, diagnostic analytics to explain why certain career paths score higher or lower based on demand drivers and risk factors, and prescriptive analytics by recommending more stable career paths and highlighting which majors students should prioritize to improve employability.

Practically, the project produces a dashboard and an index that can be reused as a decision-support framework. This makes the work valuable not only as a course deliverable but also as a model that could be extended for universities, career services, or future research on graduate employability and workforce planning.